

Multimodal Time-series Corpus

Task 1 Electricity Consumption

This is the electricity consumption recorded in Kilowatt (kW) in **city A**. Scenario: Suppose that there is a heat wave in **city A** from 2012-10-09 18:00:00 for 3 hours, leading to excessive use of air conditioning, and 5 times the usual electricity being consumed.

Task 2 Temperature Sensor

A temperature **sensor** in **city A** malfunctioned from 2025-10-01 00:00 to 2025-10-06 11:00, causing abnormal readings.

Task 3 Water Demand

Daily water demand in **city A** was much higher during the heat wave from 2012-10-09 ...

Task 4 Traffic Count

...



Phase 1: Entity Disambiguation

1

Entity Extraction

LLM

Entities

Entity	Tasks	Primary Entity
City	1, 2, 3	Task 1, Task 3

2

Profile Generation

LLM

City Profile

Name: Veridian ...

3

Profile Assignment

LLM

Rewritten Context

This is the electricity consumption recorded in kW in **Veridian city**. Suppose that there is a heat wave in **Veridian city** from 2012-10-09 18:00:00 ...

Phase 2: Multi-hop Decomposition

4

Evidence Extraction

LLM

Supporting Evidences

E1: Suppose that there is a **heat wave** in Veridian Heights from **2012-10-09 18:00:00 for 3 hours**

E2: leading to **excessive use of air conditioning**, and **5 times the usual electricity** being consumed

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Reasoning Chain Expansion

Controllable Difficulty (Reasoning hops)

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Expanded Reasoning Trace

Reasoning Evidence 1

Reasoning Link 1

Reasoning Evidence 2

Reasoning Link 2

Reasoning Evidence 3

Traceably Connected

Supporting Meta 1

Supporting Meta 2

...

Phase 3: Document Generation

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Distractor Meta Generation

Timeseries

Distractor Metadata Generation

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Distractor Meta

META

8

Meta2Doc Generation

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Supporting & Distractor Docs

Supporting Meta

META

6

Supporting Meta Generation

Timeseries

Supporting Metadata Generation

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